

INTERNATIONAL STANDARD

**Fibre optic interconnecting devices and passive components – Fibre optic
connector interfaces –
Part 20-100: Interface standard for LC connectors with protective housings
related to IEC 61076-3-106**



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INTERNATIONAL ELECTROTECHNICAL COMMISSION

**FIBRE OPTIC INTERCONNECTING DEVICES
AND PASSIVE COMPONENTS –
FIBRE OPTIC CONNECTOR INTERFACES –**
**Part 20-100: Interface standard for LC connectors
with protective housings related to IEC 61076-3-106**

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International Standard IEC 61754-20-100 has been prepared by subcommittee 86B: Fibre optic interconnecting devices and passive components, of IEC technical committee 86: Fibre optics.

The text of this standard is based on the following documents:

FDIS	Report on voting
86B/3386/FDIS	86B/3434/RVD

Full information on the voting for the approval of this standard can be found in the report on voting indicated in the above table.

This publication has been drafted in accordance with the ISO/IEC Directives, Part 2.

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- reconfirmed,
- withdrawn,
- replaced by a revised edition, or
- amended.

A bilingual version of this publication may be issued at a later date.

FIBRE OPTIC INTERCONNECTING DEVICES AND PASSIVE COMPONENTS – FIBRE OPTIC CONNECTOR INTERFACES –

Part 20-100: Interface standard for LC connectors with protective housings related to IEC 61076-3-106

1 Scope

This part of IEC 61754 covers connectors with protective housings. The housing is defined as variant 4 in IEC 61076-3-106:2006. These connectors use a push-pull coupling mechanism.

To connect the fibres inside the housing the LC interface is used as described in IEC 61754-20:2002.

The fully assembled variants (connectors) described in this document incorporate fixed and free connectors.

2 Normative references

The following documents, in whole or in part, are normatively referenced in this document and are indispensable for its application. For dated references, only the edition cited applies. For undated references, the latest edition of the referenced document (including any amendments) applies.

IEC 60050-581, *International Electrotechnical Vocabulary (IEV) – Chapter 581: Electromechanical components for electronic equipment*

IEC 60050-731, *International Electrotechnical Vocabulary (IEV) – Chapter 731: Optical fibre communication*

IEC 60529, *Degrees of protection provided by enclosures (IP Code)*

IEC 61076-3-106:2006, *Connectors for electronic equipment – Product requirements – Part 3-106: Rectangular connectors – Protective housings for use with 8-way shielded and unshielded connectors for frequencies up to 600 MHz for industrial environments incorporating the 60603-7 series interface*

IEC 61754-20:2002, *Fibre optic interconnecting devices and passive components – Fibre optic connector interfaces – Part 20: Type LC connector family*

3 Terms and definitions

For the purposes of this document, the terms and definitions given in IEC 60050-581 and IEC 60050-731 apply.

4 Description

The industrial connectors referenced in this International Standard are composed of LC-connectors according to IEC 61754-20:2002 in an IP65 and IP67 plug to receptacle interface according to IEC 60529. The intermateability is defined by the interface of the

LC-connector within the protective housing. For further details, see IEC 61754-20:2002 and IEC 61706-3-106 variant 04.

The connector housing according to IEC 61076-3-106:2006 variant 04 provides a push-pull adaptor connector interface. Inside the housing two LC connectors may be mounted to provide a duplex connection.

Dimensions are given in millimetres; drawings are shown in first angle projection. The shape of connectors may deviate from those shapes given in the following pictures as long as the specified dimensions are not influenced.

5 Interfaces

5.1 General

Connector interfaces are in accordance with IEC 61754-20:2002. This standard contains the following standard interfaces:

- Interface 11-1: duplex free plug (cable side) connector interface – PC
- Interface 11-2: duplex fixed adaptor (device side) connector interface – PC
- Interface 11-3: duplex active device receptacle interface
- Interface 11-4: duplex free plug connector interface – APC 8 °
- Interface 11-5: duplex fixed connector (adaptor) interface – APC 8 °
- Interface 11-6: duplex free plug connector interface – Large Core
- Interface 11-7: duplex fixed connector (adaptor) interface – Large Core

Table 1 shows plug to adaptor/receptacle intermateability.

Table 1 – Plug to Adaptor/Receptacle intermateability

Plugs	Adaptors/Active device receptacle interfaces			
	61754-20-100-2	61754-20-100-3	61754-20-100-5	61754-20-100-7
Interface 61754-20-100-1	Mate	Mate	Not mate	Not mate
Interface 61754-20-100-4	Not mate	Mate	Mate	Not mate
Interface 61754-20-100-6	Not mate	Mate	Not mate	Mate

Table 2 shows LC plug to free duplex plug intermateability.

Table 2 – Plug to Plug intermateability

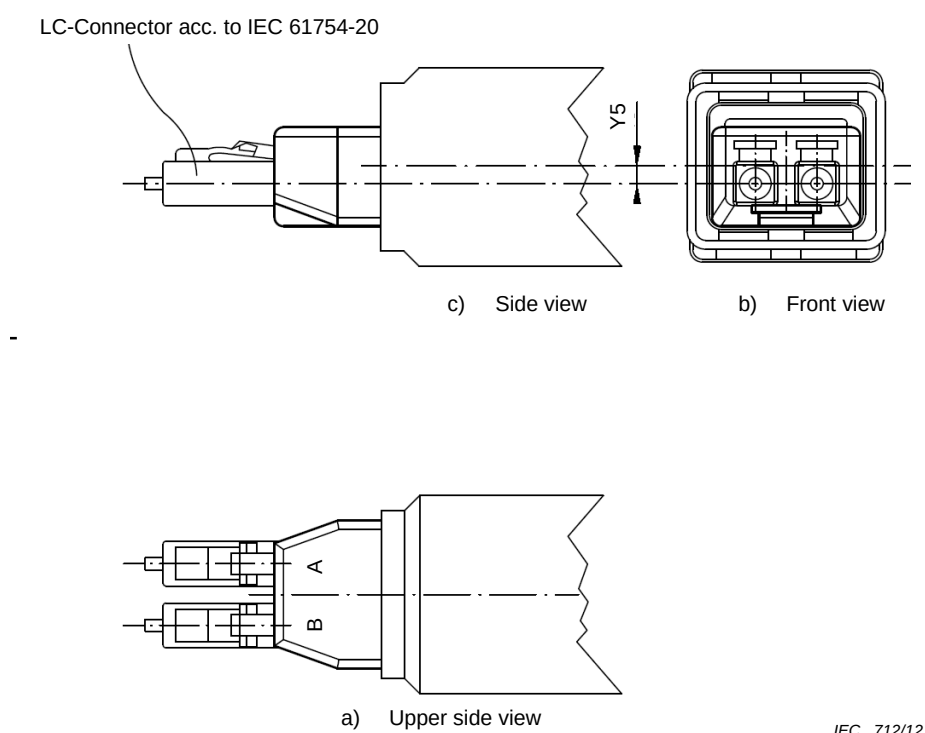
Plug (end face condition) ^a	Plug (end face condition) ^a		
	61754-20-100-1	61754-20-100-4	61754-20-100-6
Interface 61754-20-1 (PC simplex)	Mate	Not mate	Not mate
Interface 61754-20-4 (PC duplex)	Mate	Not mate	Not mate
Interface 61754-20-7 (APC simplex)	Not mate	Mate	Not mate
Interface 61754-20-8 (APC duplex)	Not mate	Mate	Not mate
Interface 61754-20:2002-simplex large core	Not mate	Not mate	Mate
Interface 61754-20-6 duplex large core	Not mate	Not mate	Mate

^a See 61754-20:2002 for plug end face conditions of PC and APC 8 °.

5.2 Industrial IEC 61754-20-100, free duplex plug connector

The "free" connector, as shown in Figure 1, is a duplex LC plug connector within a rectangular protective push-pull housing.

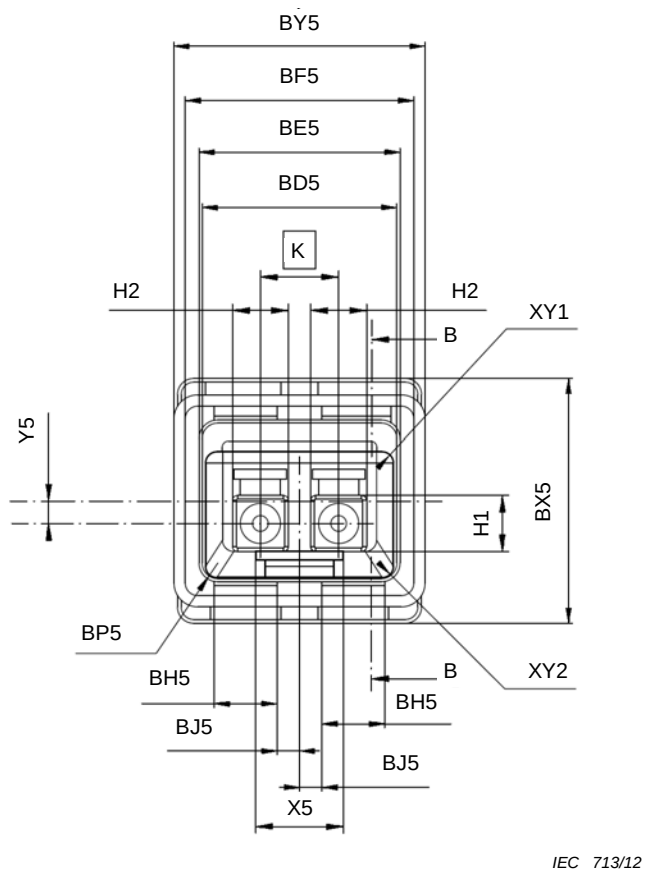
First angle projection



Key

For dimensions see Table 3

Figure 1 – Free connector

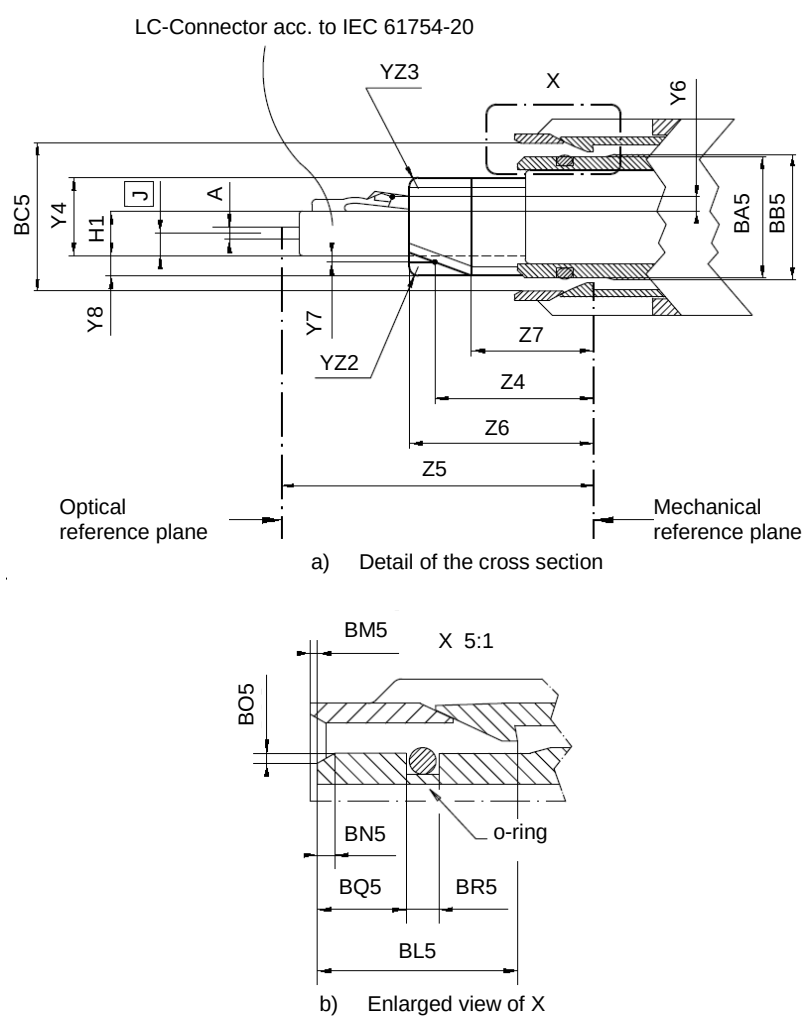


Key

For dimensions see Table 3

Figure 2 – Details of the front view

Figure 3 shows a cross section B – B.



IEC 714/12

NOTE For dimensions values of variant 04 free connector see IEC 61076-3-106:2006.

Key

For dimensions see Table 3

Figure 3 – Cross section B – B

Table 3 – Dimensions for free connector

Dimension	Minimum mm	Nominal mm	Maximum mm	Notes
BA5	12,34	12,37	12,40	^a
BB5	12,83	12,86	12,89	^a
BC5	15,10	15,15	15,20	^a
BD5	15,64	15,67	15,70	^a
BE5	16,11	16,14	16,17	^a
BF5	18,20	18,25	18,30	^a
BH5	4,95	5,00	5,05	^a
BJ5	1,75	1,80	1,85	^a
BL5	7,75	7,80	7,85	^a
BM5	0,10	0,20	0,30	^a
BN5	0,70	0,80	0,90	^a
BO5	0,70	0,75	0,80	^a
BP5	Radius 1,22	Radius 1,25	Radius 1,28	^a
BQ5	3,90	4,00	4,10	^a
BR5	1,70	1,75	1,80	^a
BX5	20,05	20,10	20,15	^a
BY5	19,70	19,85	20,00	^a
A	-	-	-	^b Diameter, Grade, Table 1b
H1	4,42		4,52	^b
H2	4,42		4,52	^b
J		H1/2		^b Basic dimension
K		6,25		^b Basic dimension
X5	-	-	6,60	Bottom shroud nose
Y4	-	-	7,98	Shrouds
Y5	1,58	1,74	1,90	Distance from ferrules centre lines to plug centre line
Y6	-	-	1,85	Lever locking height
Y7	-	-	0,67	Bottom shroud
Y8	-	-	1,87	Bottom shroud
YZ2	(Radius 0,90)	-	-	radius on bottom shroud nose(optional)
YZ3	Radius 0,90	-	-	radii on top shroud
Z4	-	-	16,28	Bottom shroud
Z5	31,76	31,93	32,10	^c Distance from ferrules optical reference plane to hood latching hooks (mechanical reference plane)

Table 3 (2 of 2)

Dimension	Minimum mm	Nominal mm	Maximum mm	Notes
Z6	-	-	18,83	Distance from shrouds front to hood latching hooks (mechanical reference plane)
Z7	-	-	12,64	Shrouds
XY1	Radius 0,90	-	-	Radii on top shroud
XY2	Radius 0,90	-	-	Radii on bottom shroud
^a From IEC 61076-3-106:2006 variant 04. ^b From IEC 61754-20:2002. ^c Dimension Z5 is given for a plug end face when not mated. The ferrule is movable by a certain axial compression force, with direct contacting end face, and therefore dimension Z5 is variable. Ferrule compression force shall be 5,0 N to 6,2 N when the position of the optical datum target, dimension Z5 is moved to the range 31,3 mm to 31,5 mm. Forces are for buffered fibre only, different cord constructions can result in higher forces.				

Table 4 – Plug connector interface – Ferrule grade

Grade	ØA mm		Notes
	Minimum	Maximum	
1	1,2485	1,2495	
2	1,2483	1,2495	
3	1,2467	1,2495	
4	-	1,2495	Large core

For all other dimensions of the housing, refer to 3.8.2 of IEC 61076-3-106:2006 (Industrial IEC 60603-7 variant 04, free connector).

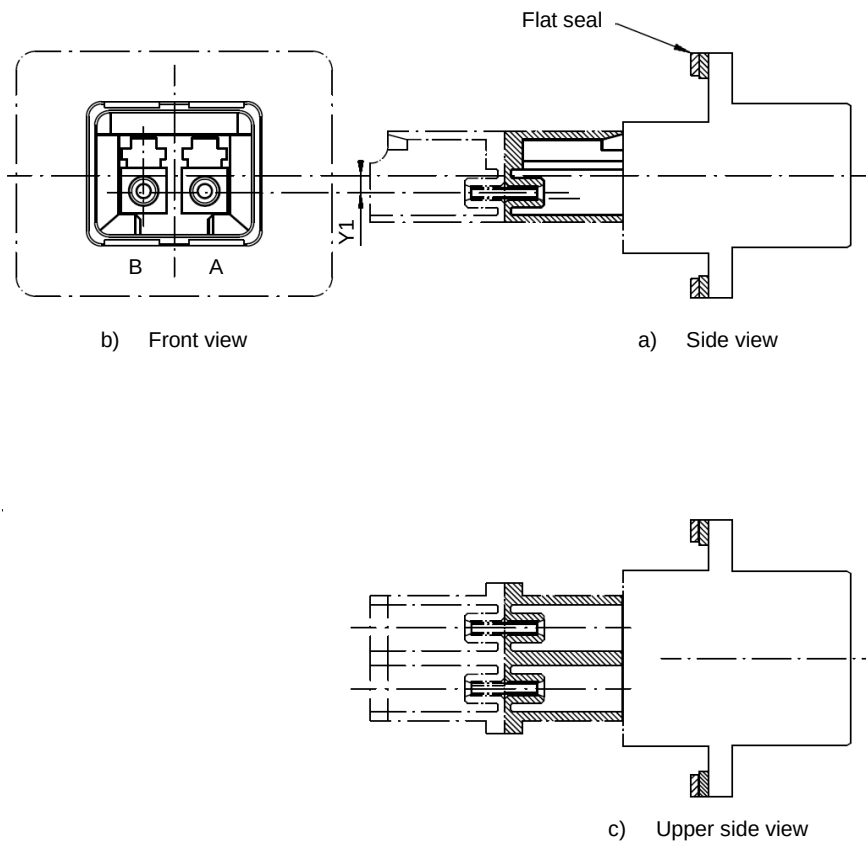
For all other dimensions of the LC-interface refer to IEC 61754-20:2002.

Terminations are according to IEC 61754-20:2002.

5.3 Industrial IEC 61754-20-100 fixed adaptor connector and active device receptacle

The "fixed" adaptor connector, as shown in Figure 4, includes a panel-mounted standard LC duplex adaptor within a protective rectangular housing equipped with a flat seal. On the inward side of the adaptor shown in phantom lines in Figure 2, the appropriate standard duplex LC connector or two standard simplex LC connectors are used.

First angle projection



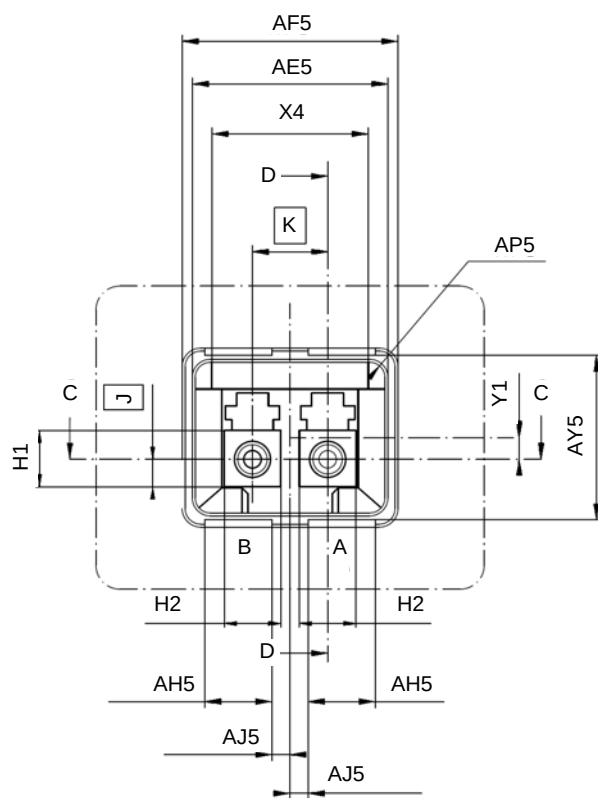
IEC 715/12

Key

For dimensions see Table 5

Figure 4 – Fixed connector and active device receptacle

Figure 5 gives details of the front view.



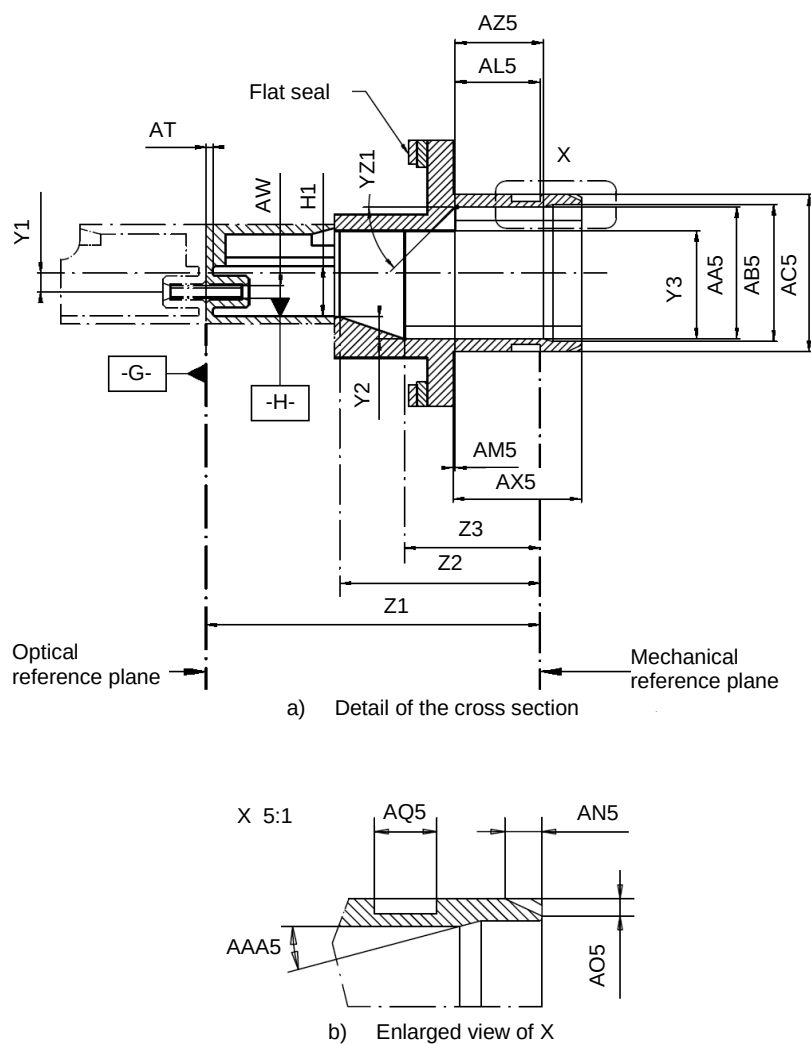
IEC 716/12

Key

For dimensions see Table 5

Figure 5 – Details of the front view

Figure 6 shows a cross-section D – D.



IEC 717/12

NOTE 1 For dimensions values of variant 04 fixed connector see IEC 61076-3-106:2006.

NOTE 2 For all other dimensions about LC Connector see IEC 61754-20:2002.

NOTE 3 Half of the duplex adaptor, or a duplex receptacle, up to the optical plane is shown.

NOTE 4 Duplex interface, same as half of duplex adaptor shown above, and IEC 61754-20:2002 Figure 7.

Key

For dimensions see Table 5

Figure 6 – Cross section D – D

Table 5 gives dimensions for a fixed connector.

Table 5 – Dimensions for fixed connector

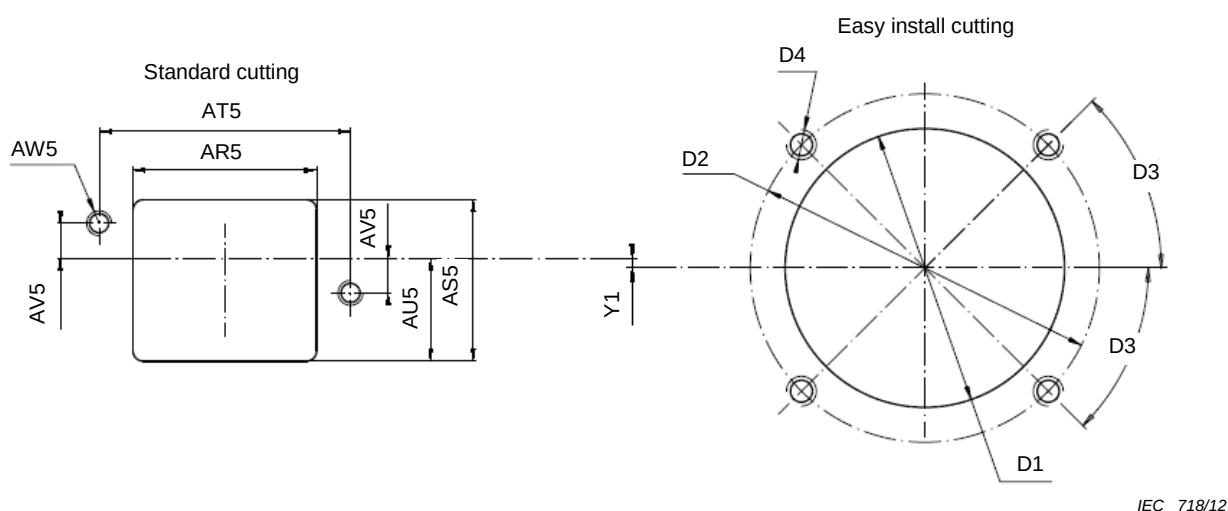
Dimension	Minimum mm	Nominal mm	Maximum mm	Notes
AA5	12,46	12,49	12,52	^a
AB5	12,95	12,98	13,01	^a
AC5	14,80	14,85	14,90	^a
AE5	16,23	16,26	16,29	^a
AF5	17,90	17,95	18,00	^a
AH5	5,50	5,60	5,70	^a
AJ5	1,45	1,50	1,55	^a
AL5	7,95	8,00	8,05	^a
AM5	0,55	0,60	0,65	^a
AN5	1,25	1,30	1,35	^a
AO5	0,55	0,60	0,65	^a
AP5	Radius 1,22	Radius 1,25	Radius 1,28	^a
AQ5	2,65	2,70	2,75	^a
AX5	12,45	12,50	12,55	^a
AY5	13,50	13,55	13,60	^a
AZ5	8,20	8,35	8,50	^a
AAA5	14 °	15 °	16 °	^a
AT	0,60		0,70	^b
AW	1,40		1,50	^b
H1	4,65		4,75	^b
H2	4,65		4,75	^b
J		2,29		^b Basic dimension
K		6,25		^b Basic dimension
X4	13,00	13,00	13,05	Stops distance
Y1	1,58	1,74	1,90	Distance from adaptor sleeves centre line to housing front centre line
Y2	2,09	2,09	2,14	Ramps
Y3	10,37	10,37	10,43	Top ramps
Z1	31,38	31,48	31,58	Distance from adaptor optical reference plane to housing latching hooks (mechanical reference plane)
Z2	18,86	18,86	18,96	Distance from housing ramps to housing latching hooks (mechanical reference plane)
Z3	12,76	12,76	12,96	Distance from housing ramps to housing latching stops (mechanical reference)
YZ1	40 °	45 °	45 °	Chamfer angle at AL5 level
^a From IEC 61076-3-106:2006 variant 04				
^b From IEC 61754-20:2002				

For all other dimensions of the housing refer to 3.8.1 of IEC 61076-3-106:2006 (Industrial IEC 60603-7 variant 04, fixed connector).

For all other dimensions of the LC-interface refer to IEC 61754-20:2002, Clause 3.

5.4 Panel cut out

The connector has a flat seal to a box or a cabinet and is accommodated by either of the following panel cut out.



Key

For dimensions see Table 6

Figure 7 – Panel cut out

Table 6 gives dimensions of the panel cut out.

Table 6 – Dimensions of panel cut out (Figure 7)

Dimension	Minimum mm	Nominal mm	Maximum mm	Notes
AR5	18,30	18,40	18,50	^a
AS5	16,10	16,20	16,30	^a
AT5	25,10	25,20	25,30	^a
AU5	10,30	10,35	10,40	^a
AV5	3,45	3,50	3,55	^a
AW5		M2,5		
Variant Easy install panel				
D1	27,80	28,00	28,20	
D2	34,90	35,00	35,10	
D3	-	45 °	-	
D4	-	M3	-	
Y1	0,90	1,00	1,10	Vertical distance between standard and easy install Cuttings
^a From IEC 61076-3-106:2006 variant 04.				

Bibliography

IEC 61754-1, *Fibre optic connector interfaces – Part 1: General and guidance*

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